Cluster Size vs E_T — Per-Period, Per-Selection

Data vs MC with truth-vertex reweighting and DI mixing

PPG12

April 16, 2026

What is plotted. Per-cluster mean tower count n_{owned} (from `cluster_ownership_array`) and 7×7-window bounding-box widths Δi_η, Δi_ϕ vs reco E_T, split per crossing angle, for three orthogonal cluster selections:

- **baseline**: fiducial $|\eta_{\text{reco}}| < 0.7$ and reco vertex $|z_r| < 60\,\text{cm}$ only.
- **common cut**: baseline plus the PPG12 common-region cut block from `config_showershape_{1p5rad,0rad}` (`cluster_prob`, $e_{11}/e_{33} < 0.98$, $\langle w_{\eta,\text{cogx}}\rangle < 2$, NPB score > 0.5).
- **tight cut**: common plus the parametric tight block (ET-dependent width and energy-fraction bounds and the `base_v1E` BDT threshold bdt $> -0.02\,E_T + 0.86$).

Sim weighting uses the data-driven iterative truth-vertex reweight $w(z_h)$ from `efficiencytool/truth_vertex_rewe` (per-period `reweight.root`; method described in `truth_vertex_reco_check.pdf`) consumed the same way `ShowerShapeCheck.C` does. Data is filtered by run range (51274–54000 for 1.5 mrad, 47289–51274 for 0 mrad). DI mixing uses $\langle y\rangle_{\text{mix}} = (1-f)\langle y\rangle_{\text{single}} + f\langle y\rangle_{\text{double}}$ applied bin-by-bin to the pooled-MC means, with $f = 0.079$ (1.5 mrad) or 0.224 (0 mrad).

Key findings.

- **The common cut removes ∼80% of the data−MC cluster-size gap**, and the tight cut essentially closes it. At $E_T = 20\,\text{GeV}$ and 1.5 mrad, baseline $\langle n_{\text{owned}}\rangle_{\text{data}} \simeq 15.2$ (jet MC 12.9, photon MC 11.0); after common it drops to $\simeq 11.6$ (jet MC 11.5, photon MC 10.2); after tight it drops to $\simeq 9.9$ essentially matching photon MC tight (10.0). The **tight-selected photon MC is the direct comparator for tight-selected data**, and the two sit within a few percent of each other across the full E_T range.
- **Both η-asymmetry and the data excess are selection-driven.** $\langle\Delta i_\eta\rangle$ shows the largest baseline data−MC gap; both the common and tight cuts collapse it to the MC band. $\langle\Delta i_\phi\rangle$ is essentially data-MC consistent at every selection stage — corroborating the March-report interpretation that the baseline excess is noise/threshold contamination preferentially stretched in η rather than a genuine shower feature.
- **The 0 mrad data shows smaller baseline cluster sizes than 1.5 mrad data for $E_T \lesssim 20\,\text{GeV}$** at the baseline and common levels, most likely due to the broader luminous region shifting more clusters to larger $|z|$ where the 7×7 window catches less of the shower's far-side tail. The tight-selected distributions at the two crossing angles converge to within statistical uncertainty — tight selection is close to period-invariant.
- **DI mixing is sub-leading in this selection.** At 1.5 mrad ($f = 7.9\%$) the DI-blend shift in $\langle n_{\text{owned}}\rangle$ is below the marker-size level for every selection. At 0 mrad ($f = 22.4\%$) the DI blend sits slightly above the pure single-MC curves in the jet pool, visible but within the per-bin statistical errors after the common cut. See the DI-specific figures (Figs. 5, 6) for the pure single vs. double MC comparison before any cut.

1.5 mrad (nominal period)

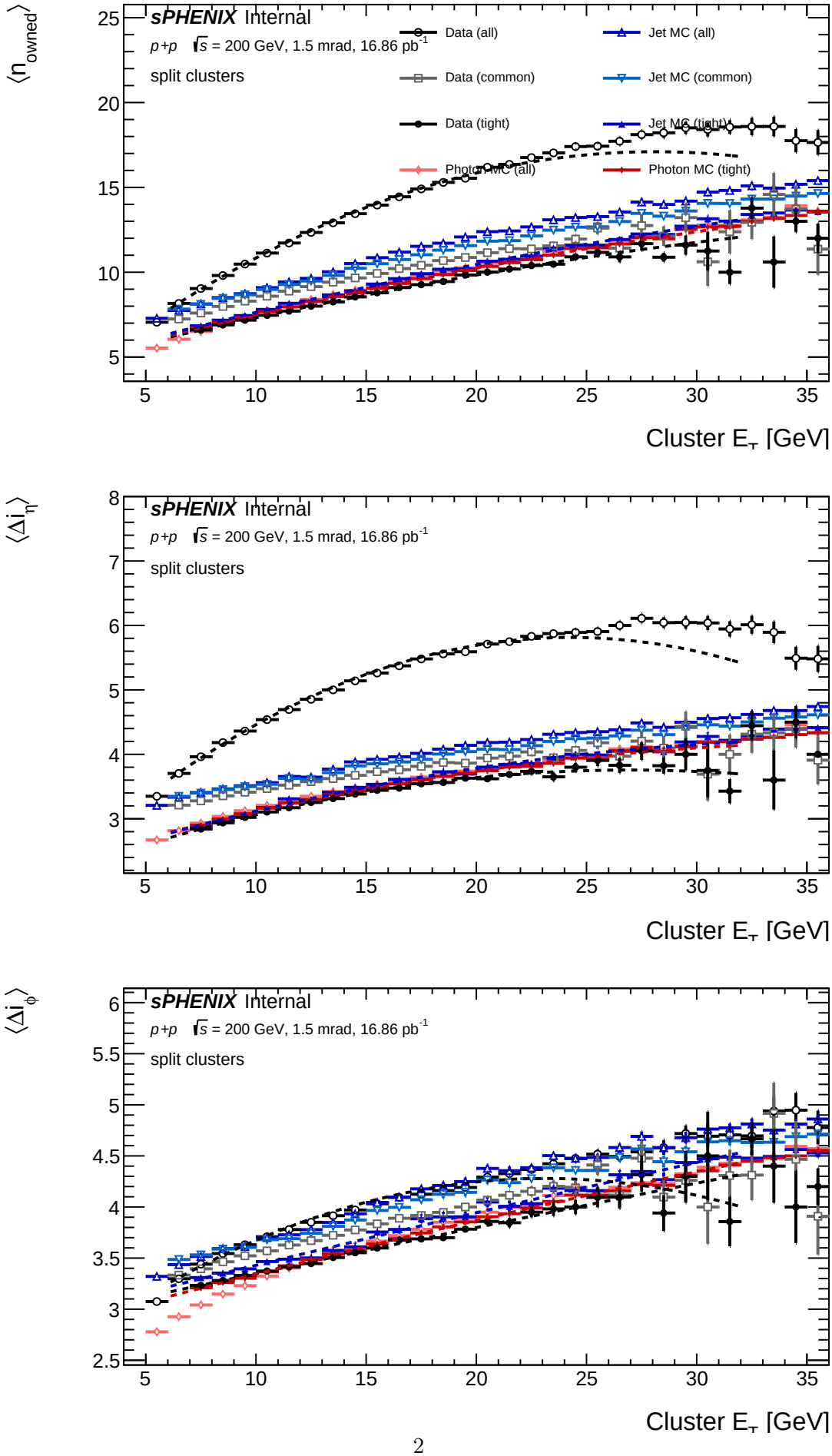

Figure 1: 1.5 mrad, split container. Overlay of data and MC at three selections: baseline (open markers), common-pass (half-emphasized), tight-pass (filled). The photon-MC tight-pass curve is the direct signal-

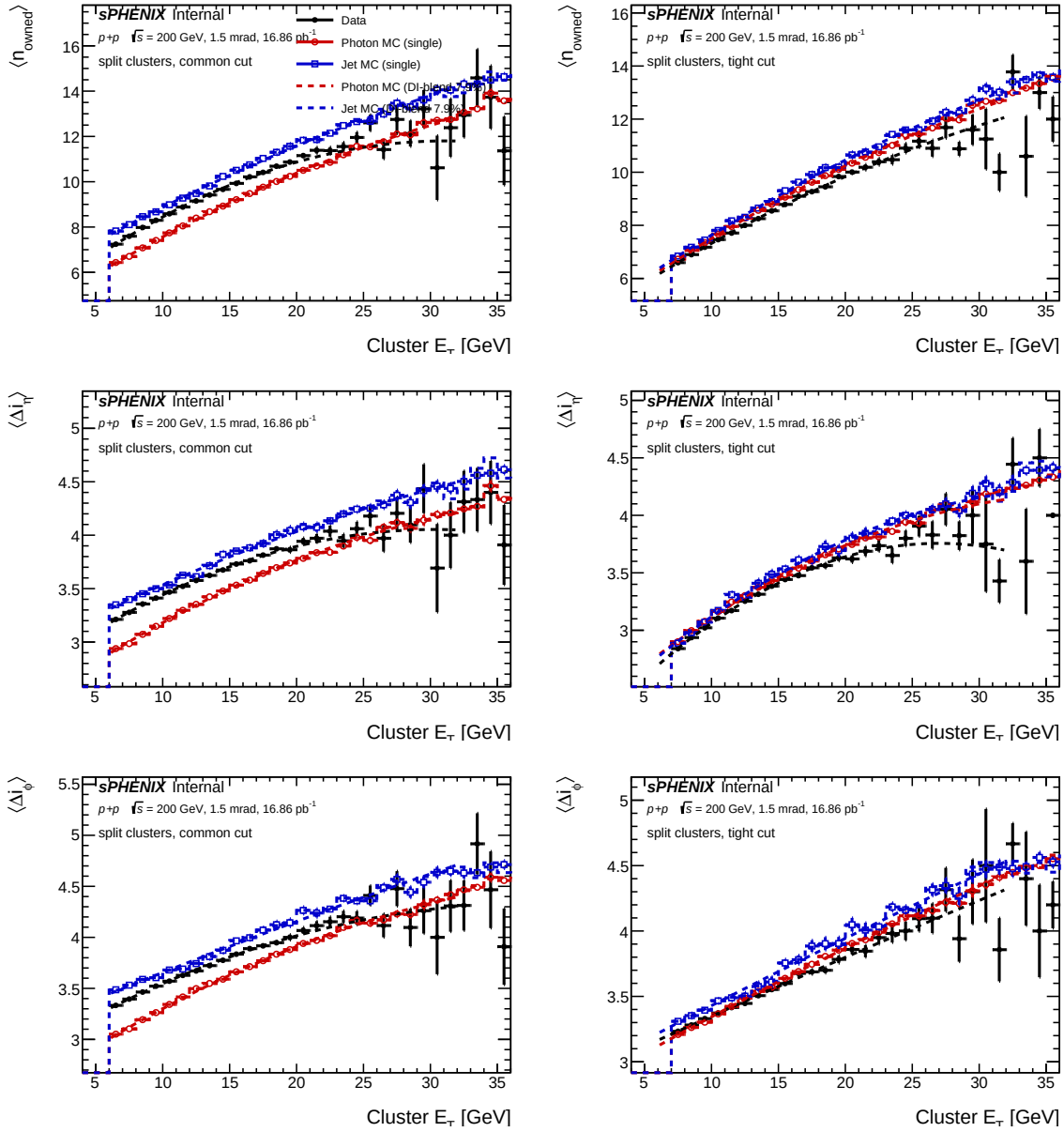

Figure 2: 1.5 mrad, split container. Three metrics (rows) × two selections (columns: common, tight), with the DI-blended MC curves drawn as dashed overlays. Same samples and weighting as Fig. 1. This view makes the MC pooling (single vs DI-blended) easy to read for each selection separately.

0 mrad (early period)

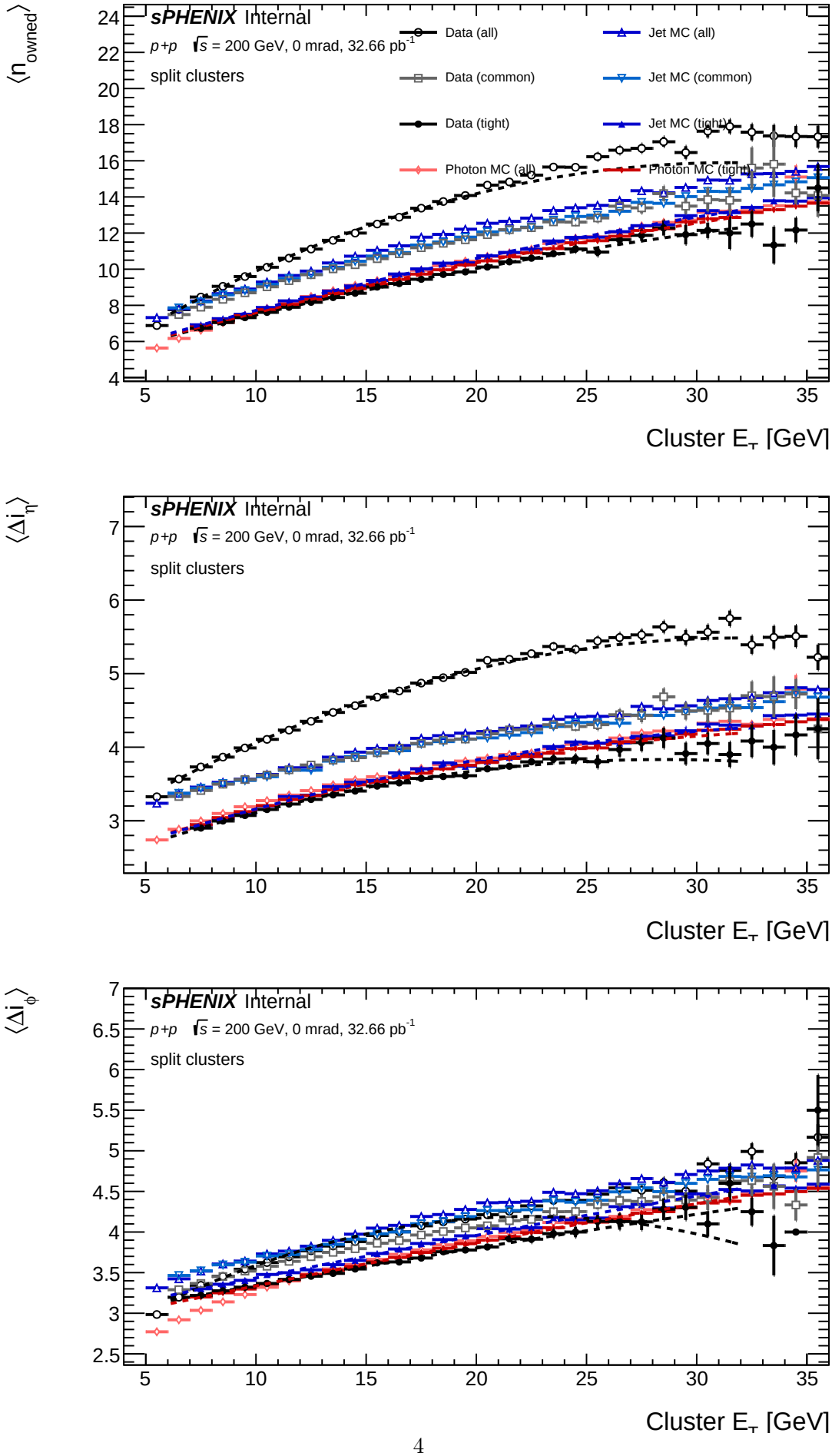

Figure 3: 0 mrad, split container. Same overlay format as Fig. 1. Lumi 32.66 pb^{-1}; truth-vertex reweight is the 0 mrad `reweight.root` (converged in 7 iterations, cf. `truth_vertex_reco_check.pdf`).

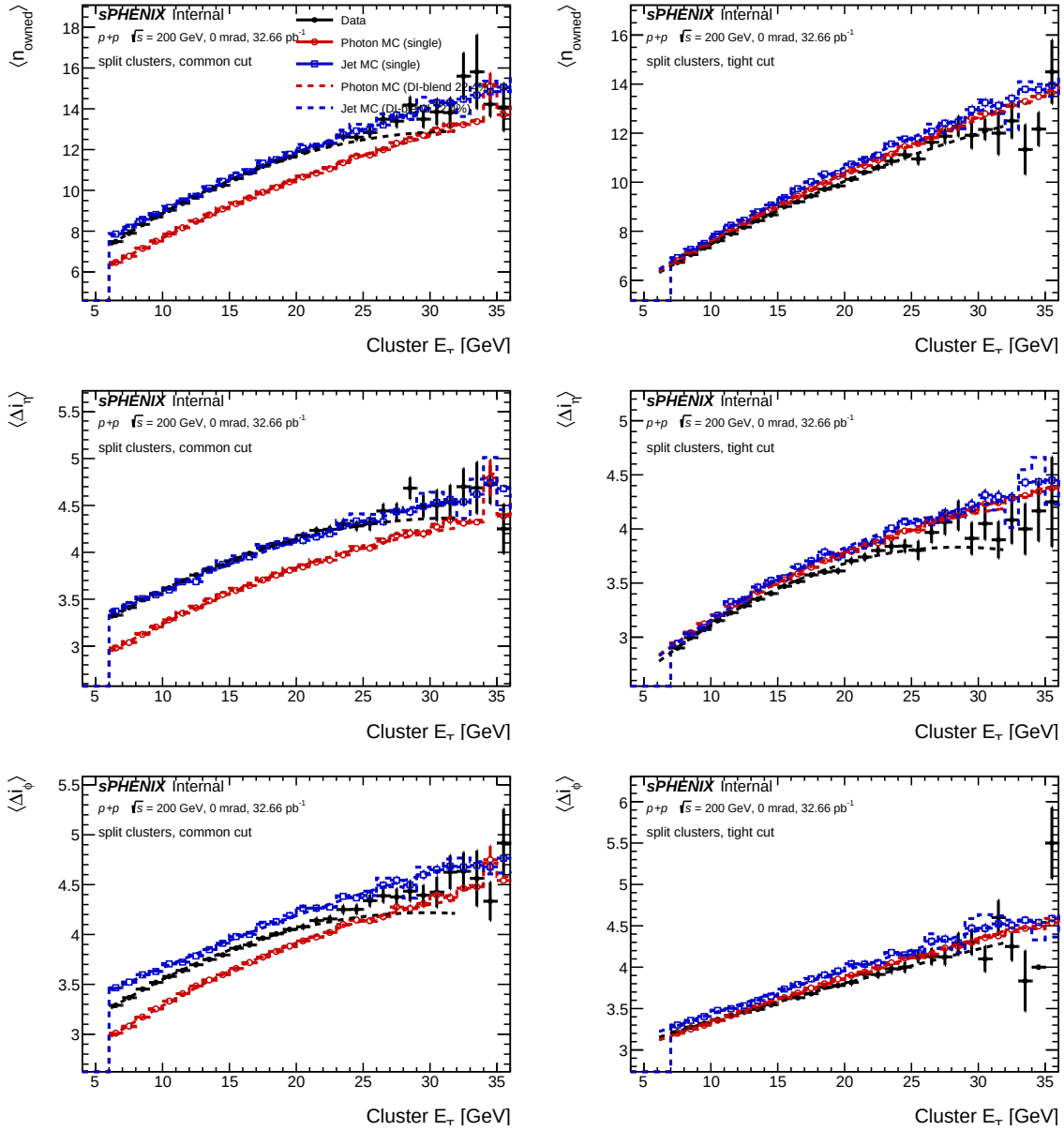

Figure 4: 0 mrad, split container. Same layout as Fig. 2 but for the 0 mrad period.

Single vs double interaction (pre-cut reference)

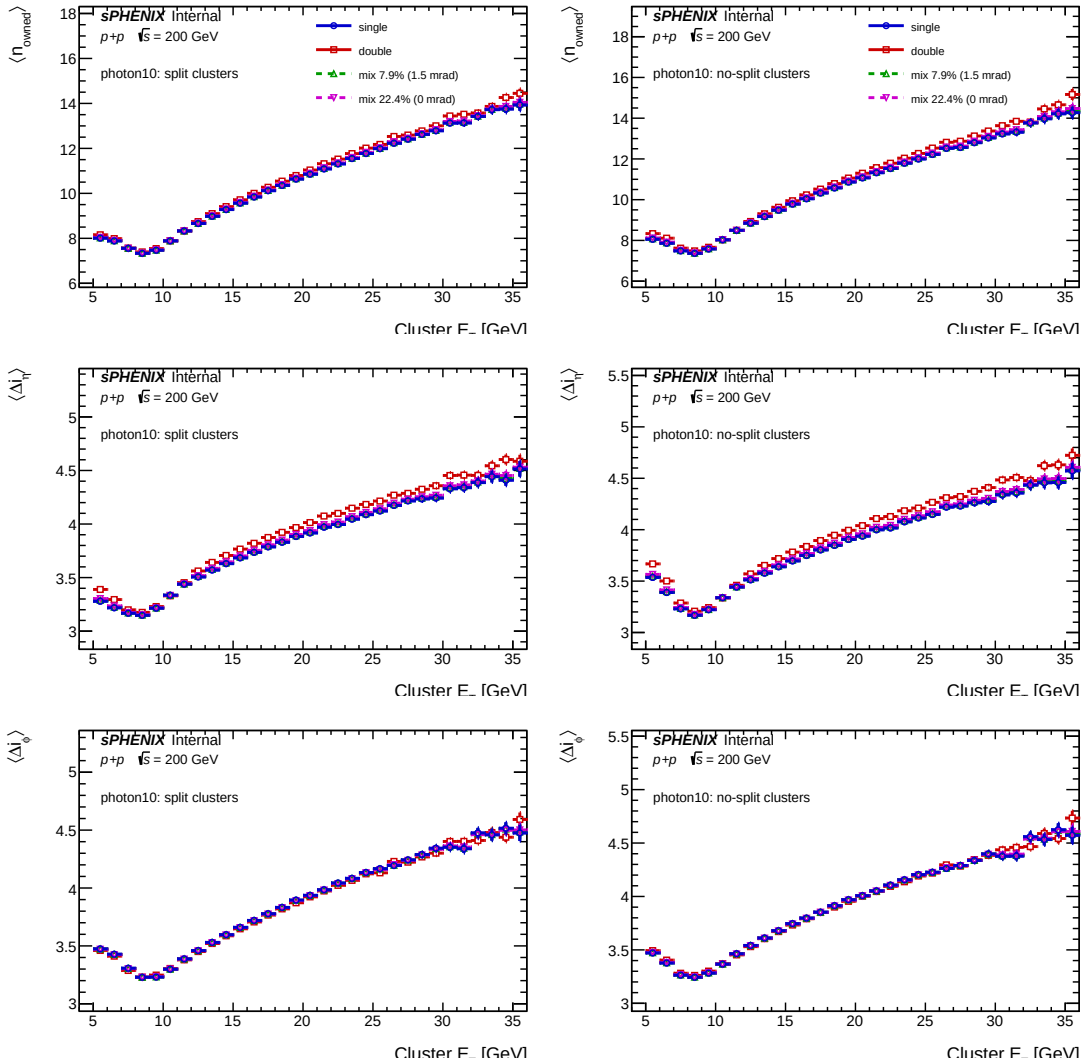

Figure 5: Photon DI reference: `photon10_nom` (blue) vs `photon10_double` (red), plus the two mixed periods ($f = 7.9\%$ green, $f = 22.4\%$ magenta). Pre-cut, both containers. This figure is unchanged from the March short report and does not apply the truth-vertex reweight — it is retained as a direct sample-level reference for the DI effect.

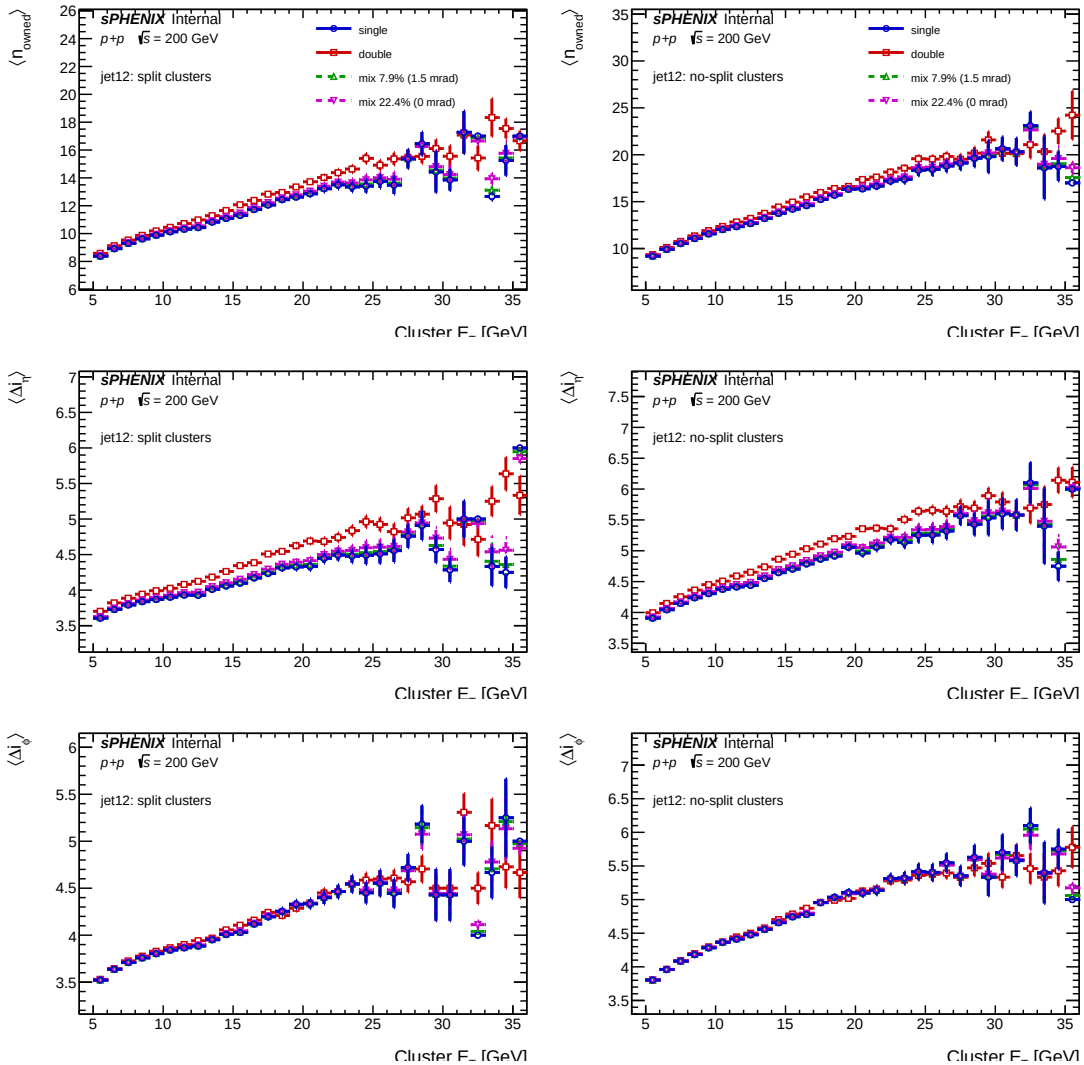

Figure 6: Jet DI reference: `jet12_nom` vs `jet12_double` in the same format as Fig. 5. DI effect on jet cluster size is several times larger than on photon, consistent with softer neighbouring towers getting pushed above the 70 MeV threshold.

Methods, caveats, reproducibility

Cuts. The common and tight blocks are loaded verbatim from the showershape configs. The tight BDT threshold uses the `base_v1E` model (the dominant ET-binned PPG12 choice) with the ET-dependent threshold bdt $> -0.02\,E_T + 0.86$ and the fixed parametric et1, $\langle w_\eta \rangle$, $\langle w_\phi \rangle$ bounds from `tight:` in the YAML. The common block uses prob $\in [0, 1]$, $e_{11}/e_{33} \in [0, 0.98]$, NPB score > 0.5, $\langle w_{\eta,\mathrm{cogx}} \rangle < 2$.

Vertex reweighting. The nominal method is the data-driven iterative truth-vertex reweight $w(z)$ consumed by `ShowerShapeCheck.C` via `truth_vertex_reweight_on: 1` in the showershape configs and the `h_w_iterative` histogram in `reweight.root`. Per event, the MC weight is multiplied by $w(z_h)$ for single-interaction MC and by $w(z_h)\,w(z_{\mathrm{mb}})$ for double-interaction MC. *Caveat*: the present `photon10_double` and `jet12_double` `bdt_split.root` files were produced before the `vertexz_truth_mb` branch was added to the anatreemaker output, so the secondary-vertex factor is silently dropped for those two samples — an optional-pointer guard identical to the one in `ShowerShapeCheck.C`. The impact on cluster-size means is expected to be below the percent level at 1.5 mrad (where the DI blend fraction is only 7.9%) and sub-percent at 0 mrad (where the MC beam profile is already close to the luminous region). Regenerating these two `bdt_split.root` files from the reprocessed `combined.root` (which does carry `vertexz_truth_mb`) is a follow-up and would remove this caveat.

DI mixing. Applied bin-by-bin at the pooled-MC level: $\langle y \rangle_{\mathrm{mix}}(E_T) = \langle y \rangle_{\mathrm{single\ pool}}(E_T) + f\,[\,\langle y \rangle_{\texttt{photon10_double}}(E_T) - \langle y \rangle_{\texttt{photon10}}(E_T)\,]$ (and analogously for the jet pool with `jet12_double`/`jet12`), which avoids biasing the

blend by the per-sample event counts. The pooled single-MC uses unit sample-weights within each class and takes the cross-section-weighted mean via `TProfile::Add`.

No-split container. The `photon10_double` and `jet12_double` production predates the dual-container `apply_BDT.C`, so the no-split container lacks BDT and NPB branches for those two samples. The cuts figures for the no-split container therefore *do not* include DI-blended MC; the baseline plots include all samples.

Reproducibility.

```
cd efficiencytool
PERIOD=1p5mrad PARALLEL=6 bash run_cluster_size.sh
PERIOD=0mrad  PARALLEL=6 bash run_cluster_size.sh
cd ../plotting
root -l -b -q 'plot_cluster_size_cuts.C+g'
```

Full narrative report: `cluster_size_vs_et.pdf` has the sample inventory, detailed per-figure observations, and noise/threshold follow-up discussion that is cross-referenced from the summary bullets above.